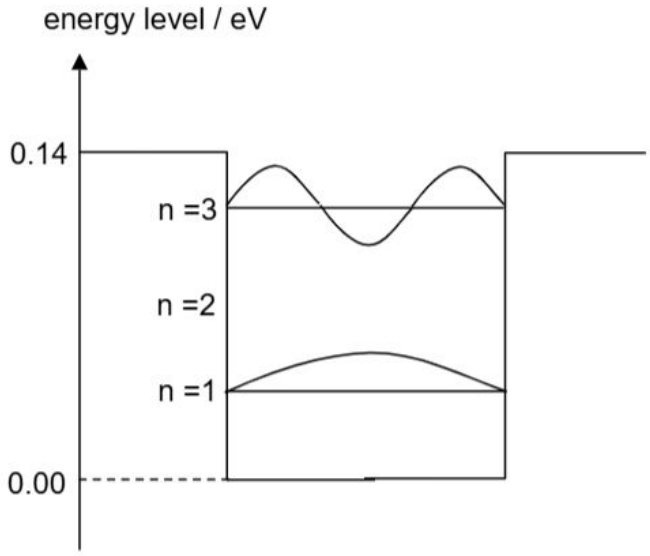
8	(a)	Number of silicon atoms per m^3: $n_{silicon} = \frac{2.33\times10^3}{0.0281}\times6.02\times10^{23} = 4.99\times10^{28}$ [C1] Number of phosphorus atoms per m^3: $n_{phosphorus} = \frac{4.99\times10^{28}}{2\times10^6}$ [M1] $=2.50\times10^{22}$ [A0]
	(b)	Using $I = nAvq$ $0.012 = 2.5\times10^{22}\times0.25\times10^{-2}\times1.1\times10^{-2}\times v\times1.6\times10^{-19}$ [M1] $v = 0.109\ \text{m s}^{-1}$ [A1]
	(c)	The electric field strength is the negative of the potential gradient. Since the potential is decreasing towards the right, or Fig. 8.3 showed negative gradient (together with electric field strength is the negative of the potential gradient), [M1] the electric field is pointing left to right. [A1]
	(d)	$E = \left\lvert\frac{\Delta V}{\Delta x}\right\rvert$ $= \frac{0.44-(-0.34)}{(0.59-0.26)\times10^{-6}}$ [M1] $= 2.36\times10^6\ \text{V m}^{-1}$ [A1]
	(e)	$C = \frac{Q}{V}$ $= \frac{3.6\times10^{-8}}{0.44-(-0.34)}$ [M1] $= 4.62\times10^{-8}\ \text{F}$ [A1]
	(f)	minimum wavelength: $\lambda = \frac{hc}{(0.29+1.42+0.34)e}$ [M1] $= 606\times10^{-9}\ \text{m}$ [A1]
	(g)	red/orange light for 606 nm [B1] infrared for 875 nm [allow ecf from part (e)]

(h) (i)



[B1] for n = 1

[B1] for n = 3

(ii)

kinetic energy $E = \frac{p^2}{2m}$ [C1]

$$E = \frac{\left(\frac{nh}{2d}\right)^2}{2m}$$ [M1]

$$E = \frac{n^2h^2}{8md^2}$$ [A0]

(iii)

When n = 3

$$E = \frac{(3)^2\left(6.67\times10^{-34}\right)^2}{8\times9.11\times10^{-31}\times\left(5\times10^{-9}\right)^2}$$ [M1]

$$= 2.17\times10^{-20}$$

$$= 0.136 \text{ eV}$$ [A1]

This is very close to the potential height of the quantum well of 0.14 eV.
The next energy level is greater, almost double due to the coefficient of 4^2 and so will exceed the energy barrier, and the electron with that amount of energy will not be confined in the well. [B1]